
Mechanistic Circuits and Concept Geometry in an Instruction-Tuned Language Model

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Motivation and Scientific Justification

Large language models can produce correct answers to arithmetic and reasoning problems without necessarily implementing the intended computation. Output accuracy therefore cannot distinguish robust internal algorithms from surface heuristics, memorised templates, or prompt-specific associations. This thesis studies these aspects in **Qwen3-4B-Instruct**, an independently trained instruction-tuned model, using mechanistic interpretability methods that analyse the residual stream, sparse transcoder features, attribution graphs, activation patching, and learned input-side interventions.

The work has three purposes. The first is to test whether circuits reported in Anthropic’s *On the Biology of a Large Language Model* [1] transfer to a different instruction-tuned model family. The second is to develop a general method for locating abstract concepts without relying on a prior circuit hypothesis. The third is to examine whether learned continuous prefixes can redirect model behaviour and whether the learned directions are interpretable relative to the concepts they represent.

Methodology

The reproducibility study applies the attribution graph framework to Qwen3-4B through a linearised transcoder model. For two digit addition, sparse features are scanned across operand grids and token positions to identify range, residue, lookup, and sum-sensitive activations. Teacher-forced accuracy and entropy analyses separate errors in the first generated answer token from later continuation effects. A second reproduction studies multilingual antonym completion by suppressing and injecting transcoder features corresponding to operation, operand, and output language.

The concept geometry study introduces a contrastive residual-stream method. For each concept, matched positive and negative prompt pairs are constructed so that syntax, length, scale, and answer format are held approximately constant. The mean residual-stream difference between the two sets defines a layer-wise concept delta at candidate token anchors. Delta norms are normalised by the residual activation scale and compared with a permutation null distribution. Inter-layer cosine similarities then show whether the concept direction remains stable across depth or rotates. Activation patching tests causal sufficiency by inserting positive-example residual states into negative examples and measuring the change in first-token output. Finally, each delta is projected through transcoder encoder and decoder directions to identify sparse features that read or write along the concept direction.

This pipeline is applied to three arithmetic contrasts of increasing representational difficulty, namely carry detection in addition, divisibility by seven through the greatest common

divisor, and residue-class membership modulo seven. The soft-prompting chapter trains ten learned prefix embeddings with base model weights frozen, evaluating whether input-side perturbations can correct residue-class completion, balanced-parentheses judgements, and causal direction judgements.

Key Findings

The addition reproduction finds the same high-level digit-pair organisation reported for Claude 3.5 Haiku. Qwen3-4B reads operand information at digit positions and recombines decimal residues at the delimiter through lookup-like and sum-sensitive features. The first answer token is the main source of difficulty, while later answer tokens become highly confident under teacher forcing. This supports a lookup-like account of the tested two digit addition format rather than a sequential carry-propagation account.

The multilingual reproduction shows that semantic operation and operand components can be edited across English, Chinese, and French. Suppressing antonym features and injecting synonym features moves completions toward synonym answers, and changing the operand concept from *small* to *hot* transfers strongly across languages. In contrast, the attempted output-language edit is weak or absent. Thus some mechanistic components transfer across languages and model families, while others depend more strongly on tokenisation, selected layer range, or prompt surface form.

The concept-localisation results show that concept geometry is not uniform. For the addition task, stable direction appears around layers 4 and 5 at the final operand token, survives comparison with the permutation null baseline, and is causally sufficient under activation patching. The aligned transcoder features include operand comparison, iso-sum, parity, and Fourier-like grid structure. GCD divisibility is also localisable, but its inter-layer cosine matrix shows a block transition, indicating that the direction rotates rather than persisting as a single axis. At the rotation layer, we see divisibility detection patterns, directly correlated with the task. Residue-class membership does not produce a stable direction above the null band, consistent with its low first-token accuracy and suggesting that this modular relation is not encoded as a linearly accessible binary contrast.

pattern with a concrete example.

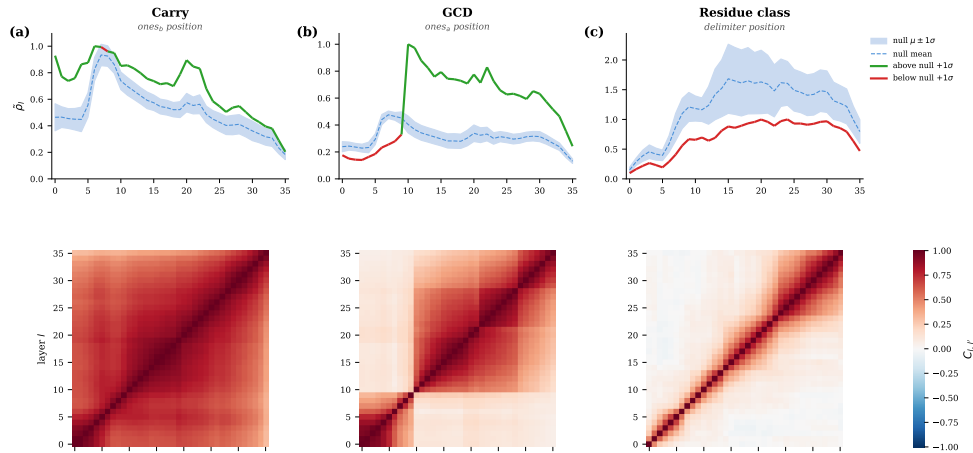


Figure 1: Contrastive residual-stream geometry for carry, GCD divisibility, and residue-class membership. Carry shows a stable direction, GCD shows a representational rotation, and residue class remains weak relative to the null baseline.

The soft-prompting experiments show that learned prefixes can redirect outputs when the main failure is response format. Accuracy rises from 0% to 28.6% on balanced parentheses and from 0% to 53.8% on causal direction judgements. Residue-class completion changes only from 15.0% to 16.0%, while the mean probability of the correct token decreases. The learned embeddings do not align with identified concept directions and are not interpretable through transcoder features, which makes soft prompting effective yet a non-interpretable transparent knowledge-editing method for our task.

Recommendations and Next Steps

The main conclusion is that mechanistic structure identified in one model family can transfer to independently trained instruction-tuned models, but that the geometry of a target concept strongly conditions whether it can be localised, patched, or edited. Stable low-rank directions, rotating directions, and absent linear contrasts require different intervention strategies. This aspect has practical implications for activation steering, probing, and knowledge editing, where the existence of an accessible representation should be established before an edit is attempted.

Future work should test whether the transcoder features identified by delta projection form causally sufficient circuits when combined with full attribution graphs. The feature ranking procedure should also account for the non-orthogonality of decoder vectors and co-activation geometry. Finally, the contrastive delta method should be applied beyond binary concepts to multi-class and continuous concepts, where the geometry may be more complex and the null distribution may need to be constructed differently.

Research Impact and Conclusion

Overall, this thesis contributes a reproduction of large-model circuit-tracing results in Qwen3-4B, a contrastive method for analysing concept geometry across layers and token positions, and an empirical evaluation of soft prompting as a residual-stream intervention. Together, these results show that instruction-tuned models can contain sparse, transferable, and interpretable computational structure, but also that correct or improved outputs do not imply transparent internal computation. Mechanistic evaluation must therefore ask where a representation is encoded, whether it is causally active, and whether an intervention changes the computation itself or only redirects the surface form of the answer.

References

- [1] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Shan Carter, Tom Henighan, Adam Stevenson, Tom Conerly, Danny Hernandez, Adam Jermy, Callum McDougall, et al. On the biology of a large language model. 2025. Anthropic Transformer Circuits Thread.